Sharan Majumder
smajumder1@ufl.edu — majumdersharan.github.io

Education

University of Florida Gainesville, FL
Ph.D in Computer Science *Expected May 2029*
B.S. in Computer Science and Mathematics *May 2025*
Relevant Coursework: Data Structures & Algorithms, Computational Geometry, Machine Learning, A.I. Fundamentals, Natural Language Processing, Maths for Intelligent Systems

Technical Skills

Programming Languages: C++, Python, Java, JavaScript, MATLAB, SQL, R, Erlang

Frameworks & Tools: GitHub, Node.js, Android Studio, Jupyter Notebook, HuggingFace, XGBoost

ML & Data: NumPy, Matplotlib, Pandas, Seaborn, Scikit-learn, PyTorch, TensorFlow, Neural Networks, Transformers, CNNs, LLMs, RAG, SHAP, LIME

Research & Work Experience

Teaching Assistant - University of Florida *August 2025 - Present*
- Assisted over 300 students by hosting weekly office hours to provide academic support and assist with topics of algorithm design, complexity analysis, and programming language concepts
- Provided constructive feedback on code quality, quizzes, and exams to improve their understanding and performance
- Setup additional study sessions prior to quizzes/exams to better assist students in the course

Simple Polygonal-Base City Guarding and Analysis of Bad Regions *January 2025 - Present*
- Investigated extensions of the Art Gallery Problem to three-dimensional city-guarding environments with simple-polygonal building bases and bounded-field-of-view cameras
- Proved sufficient camera bounds for guarding building roofs, walls, ground, and aerial free space under restricted rooftop-corner placement and orthogonal camera orientations

Impact of A.I. in Large Classroom Setting - Accepted *January 2024 - November 2024*
- Conducted a study to measure if A.I. technologies can increase student engagement in large classroom settings
- Performed data and conversation analysis of over 200 students' conversations to observe how well students were engaged in learning Observed students reported higher satisfaction and engagement compared to control group
- Submitted a paper which details the background on AI in education, how the study was conducted, and the significance of the data

Technical Projects

Pastry-Inspired Peer-to-Peer Simulator (Erlang) *October 2026 - October 2026*
- Built an actor-model simulator with one Erlang process per peer, 128-bit node IDs, prefix-based routing tables, and leaf sets
- Implemented peer joining, string key-value storage and retrieval, and configurable per-peer requests; measured overlay hop counts for message delivery

Scratch Implementation of a Feedforward Neural Network March 2026 - April 2026
for Binary Classification (Python, Numpy, Matplotlib)
- Achieved 96% classification accuracy on a dataset of 1000 points by building and training a custom neural network (2–3–1 architecture) from scratch using sigmoid activation and backpropagation over 10,000 epochs
- Developed a binary classifier to detect whether 2D points lie within a triangular region, improving predictive performance by manually implementing forward propagation, loss computation, and gradient-based weight updates
- Evaluated model generalization by expanding the input space beyond the training distribution, identifying a drop to 74% accuracy and analyzing limitations in learned decision boundaries

Signify ASL Interpreter and Study Model (Android Studio, Kotlin, SQL) *January 2025 - April 2025*
- Collaborated in a team to design and develop a mobile application for learning American Sign Language (ASL)
- Designed an implemented an SQLite relational database schema storing and querying user information, achievements, and discussions
- Enabled seamless app updates by managing database version control through onUpgrade(), ensuring new columns could be added without disrupting user data
- Increased development efficiency by creating a debug page within the app that allowed the team to test database functionality before implementation, reducing integration errors

Leadership and Professional Development

UF National Society of Black Engineers: Executive Board *May 2024 – May 2025*
Secretary
- Coordinated monthly chapter meetings and managed internal communications to ensure smooth information flow across the organization
- Delegated tasks to members for event promotion and engagement (flyer creation, photography, website updates, and social media posts)

Awards

BRIDGE Scholarship Recipient, Cum Laude Graduate, Meritorious Award SCUDEM 2024, Successful Participant MCM/ICM 2024, Dean's List Spring 2023, Successful Participant MCM/ICM 2023, Meritorious Award SCUDEM 2022